

ICML

2026

REPRODUCTION

Distribution-valued Change Point Detection: Reproduction Audit

CPU

AUDIT

OPEN

Source-pinned CPU evidence separates scoped theorem support from literal claim discrepancies.

DineshAI reproduction campaign

Independent reproduction record

1 Outcome

Six live claims were audited against the pinned paper source and official release. One scoped theorem audit is verified; four literal source claims are falsified; the tangent-space mechanism remains inconclusive with toy evidence.

2 Tangent-space mapping

Three reduced CPU transport audits pass translation, scale, and deformation identities with expected negative controls. The source-faithful R/funcharts mFPCA path is unavailable, so Claim 1 is not called verified.

3 False-alarm bound

The fixed-threshold theorem supports its displayed bound. The empirical-quantile corollary adds the finite-sample term $2/(n_0+1)$; omitting it falsifies the literal live wording.

4 Finite-dimensional theorem

Theorem 3.14 gives K proportional to $\epsilon^{(-2d)}$ under the stated regularity chain. Nine finite- K algebra checks pass; a dimension-free control fails.

5 Synthetic Table 1

Pinned high-variance rows give a maximum 72.5% reduction against the best displayed Log-KDE, not 95%. Released generator inputs ran; full R/mFPCA table execution remains unavailable.

6 FlowCAP-II

The public release lacks FlowCAP data, labels, and runner. The live claim changes Hotelling precision to F1 and conflicts with appendix ARL1 result rather than its headline description.

7 Reddit timeline

The authoritative Figure-4 source reports an April 30 post-gap reorganization after an April 3--28 sparse-data gap, not a direct April 13 alarm.

8 Reproducibility

All scripts, tests, source pins, raw logs, and hashes are public. No HF Job or Bucket was used; unavailable source artifacts are documented instead of replaced with proxies.

- Public code at the pinned commit.
- Exact live claims retained with hashes.